

strategies. On the other side of the issue, Fang, Jacobsen, and Qin (2013) found poor out-of-sample performance when they retested the exact same 26 Brock et al. trading rules on 25 years of new data. In their paper “What Do We Know About the Profitability of Technical Analysis?” Park and Irwin (2007) gave a summary of the mixed state of trend-following results. The authors found that among 95 modern studies of technical trading strategies from 1988 through 2004, there were 56 with positive results, 29 with negative results, and 19 with mixed results. The authors concluded that most empirical studies were subject to problems in their testing procedures related to data snooping, ex-post selection of rules, and not accounting properly for risks and transaction costs.

More recently, Bajgrowicz and Scaillet (2012) applied 7,846 trading rules to the daily Dow Jones Industrial Average from 1897 through 2011. Using an up-to-date correction for data-snooping bias, the false discovery rate (FDR), the authors found that investors would never have been able to select the best performance rules ahead of time.⁵ Introducing transaction costs would have eliminated whatever profits did exist.

Along the same lines, Fang, Qin, and Jacobson (2014) examined the profitability of 93 market indicators (50 market sentiment and 43 market strength indicators) applied to S&P 500 data averaging 54 years in length. After accounting for transaction costs, none of the indicators outperformed buying and holding the S&P 500 on a risk-adjusted basis, and there was no evidence that they showed predictability with respect to future stock returns.

We see that data mining for new trading rules is a perilous undertaking. Although bootstrapping can help establish confidence levels, especially when dealing with modest amounts of data, it may depend on assumptions about how the markets function that may not be realistic. More specifically, accurate time-series bootstrapping and simulation results/depend on data ergodicity and stationarity. Since financial markets are nonstationary and nonergodic with possible price jumps and nonfinite variances, bootstrap simulations may have their own potential problems.

The simplicity and robustness of absolute momentum gives it an edge over unproven and uncertain alternative trend-following methods. With that in mind, let us look cautiously at a few other trend-following approaches that are in keeping with what we know about absolute momentum.